

Insights From Longitudinal Evaluation of *Moodie* Mental Health App

Alaa Alslaity*

Faculty of Computer Science, Dalhousie University,
Halifax, Nova Scotia
Alaa.alslaity@dal.ca

Rita Orji

Faculty of Computer Science, Dalhousie University,
Halifax, Nova Scotia
Rita.Orji@dal.ca

Gerry Chan

Faculty of Computer Science, Dalhousie University,
Halifax, Nova Scotia
gerry.chan@dal.ca

Richard Wilson

Feeling Moodie, Moncton, New Brunswick
rw@feelingmoodie.com

ABSTRACT

Recently, mental health applications have gained increasing attention. This is because research shows that users find mental health apps as a good alternative for self-management of mental conditions, especially in the last two years when access to physicians was limited because of the pandemic. Despite the existence of several mobile apps for mental health, most available apps are not supported by empirical and scientific evidence, or they are designed based on experimental rather than real data. This work represents our first step toward designing evidence-based AI-driven mental health applications. In this paper, we show our initial results and discussion of in-the-wild users' interactions and engagement with a mental health app (called Moodie) for a period of two years. Specifically, we investigate the interactions and engagements of 434 users (who used the app for two years) with the mood tracking feature where the app allows users to enter their moods and the corresponding factors associated with the mood. Chi-square analysis showed a significant correlation between moods and factors (or user's activities). The results also show that Home, Work, Relaxation, and Family-related activities are the most common factors that affect moods either positively or negatively and that these factors have a different impact on moods during different times of the year.

CCS CONCEPTS

• **Human-centered computing**; • **Ubiquitous and mobile computing**; • **Empirical studies in ubiquitous and mobile computing**;

KEYWORDS

Mental health, Mobile-health app, mHealth, mood tracker, mobile app

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1 INTRODUCTION

Mental health issues have become a major public health challenge. According to the World Health Organization (WHO), one in four people in the world will experience mental disorders in their lifetime [29]. Health care services were traditionally provided through health care facilities, such as hospitals and clinics. However, there have been tremendous advances in technology that have allowed for the delivery of health care on a larger scale and more efficiently. Mobile health (mHealth) apps have shown strong evidence in supporting conventional health systems by augmenting diagnosing and treatment processes and increasing individuals' participation in managing their mental health. Due to its importance, designers and researchers have paid more attention to mental health apps and started investigating how mHealth apps can be integrated into healthcare models. According to Aitken and Lyle [23], mental health apps comprise about one-third of disease-specific apps. Despite the enthusiasm of individuals and health care authorities for mental health apps, several questions are yet to be answered. Common issues are that most of the available mental health apps do not have feasibility data based on real-world use over a long time period, lack the advisory of a health care professional, or are not evidence-based and direct research support [24].

This work is part of ongoing research towards developing an adaptive and evidence-based mental health app. In this paper, we discuss the initial results from users' interactions and engagement with a mental health app called *FeelingMoodie* (or *Moodie*) [17]. The app, which was developed in collaboration between mental health clinicians/practitioners and tech experts, provides data on moods and factors associated with these moods. It allows mental health providers to obtain significant historical data to facilitate accurate and speedy diagnosis of mental issues. Identifying the relationship between moods, environmental, physical symptoms, and self-reporting descriptive representation of a patient can help understand what factors (negative and positive) and patterns impact people's mood and well-being. As a first step in designing the

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adaptive AI-based adaptive mental health app, this work aims to explore the relationship between moods and factors and how this varies across various times and seasons (Winter, Spring, Summer, and Fall) to inform the design of better mental health apps.

To achieve our goal, we analyzed data from 434 users who used the first version of the app over the course of two years (January 2020 to December 2021). The quantitative analysis of the data revealed that Home-, Work-, Relaxation-, and Family-related activities are the most common factors that affect users' moods, while Chill emerged as the most common mood. The results also show evidence that factors have different impacts on moods during different times and seasons of the year.

The current work adds to the ongoing efforts to understand how technology can be used to enhance positive well-being and how humans interact with mood-tracking apps for improving mental wellness. The contribution of this research is three-fold: 1) it provides a longitudinal in-the-wild evaluation of user interactions of a mental health app, 2) it identifies positive and negative factors that affect users' moods and how this varies across different times and seasons, and 3) it provides insights and recommendations for designing an adaptive user-centric mental health app.

2 RELATED WORK

Mood-tracking apps have shown potential success in helping users build positive emotions and control their moods [9]. Recent studies stated that people use mood-tracking apps for two reasons: experiencing an adverse life event and changes in their mental health [26]. The effectiveness of mHealth apps is still unclear. For instance, some research suggests that mood-tracking apps can reduce the intensity of negative moods and increase the intensity of positive moods [19], while other researchers have found contradictory results [7, 16]. Accordingly, conducting longitudinal studies for mHealth apps helps researchers and health practitioners investigate the long-term effects of mHealth apps on individuals' moods.

Several researchers have conducted longitudinal studies for mHealth apps. For example, an eight-week study was conducted by Burns et al. [10] to evaluate the feasibility and satisfaction level of "*Mobilize!*", a mHealth app targeting depressive disorders. The study showed that the app was effective for reducing depression. However, the study was very limited in terms of the number of participants. Another retrospective longitudinal study by Huberty et al. [15] evaluated an app called "*Calm App*." The analysis revealed that the mood check-ins feature can increase the level of meditation and enhance users' experience with meditation. Based on the results, the authors recommend integrating mood check-in features to engage users in mHealth apps for meditation apps.

Bakker and Rickard [5] explored the relationships between app engagement and mental health outcomes regarding depression, anxiety, and mental wellbeing. The study considered a self-monitoring app called "*MoodPrism*". The results show the app's success in increasing engagement for mental wellbeing, but not for depression and anxiety. In a more recent study, Bakker and Rickard [6] investigated the relationship between mental health outcomes and engagement on another app called "*MoodMission*". The results show that apps that provide cognitive behavioural therapy strategies can increase users' engagement and enhance their mental wellbeing.

Bardem et al. [7] developed and evaluated the *MONARCA*, a personal monitoring system for bipolar patients. A 14-week field trial was conducted where 12 patients used the system. The results show that the system was successful in improving adherence to self-assessment. In a more recent study,

Despite the several studies on the effectiveness of mHealth apps, most studies are limited in the number of participants and study period, and there is a lack of research on how moods and activities are related over time. This paper analyzes data from 434 users of the first version of a mental health app over the course of two years.

3 THE MENTAL HEALTH APP

The *Feeling Moodie* (or *Moodie*) app [17] is a mental health app designed in compliance with Health Insurance Portability and Accountability (HIPAA) [4] and the Personal Information Protection and Electronic Documents Act (PIPEDA) [28]. *Moodie* app tracks mood and health-related information. The app allows users to track their moods by identifying their moods and associated factors (Home, Work, Relaxation, Family, School, Friend, Pet, Sport, Movie, Event, Game, and Party). Four moods were identified in the app: Great (Happiness or good mood), Chill (being in a calm state, or to relax completely or being relaxed), Meh (indifference, boredom, or a lack of enthusiasm), and Sad (feelings of unhappiness and low mood). These four moods were selected as general moods representing a wide range of sub-moods. It also allows users to track their emotions, moods, routines, sleep, diet, physical activity, and associated context via journal logs. Through tracking various activities via journaling and associated emotions, the app can identify patterns and activities that influence a users' mood and behaviours using machine learning.

Moodie app has four main features: (1) mood tracking where users can enter their moods, (2) Mind Fitness Plan, which provides assistant for users to change their mood to a better state using cognitive behavioral therapy [12] treatment, (3) Journaling, which allows users to track events and situations that might affect their mood and (4) Visualization where users can see historical data and track their moods, which help users' and health care practitioners to recognize emerging trends and respond rapidly. Figure 1 depicts screenshots of the *Moodie* app.

4 METHOD

We analyzed data from 434 users who used the *Moodie* app from January 2020 to December 2021. Users' records involved data about mood (Meh, Chill, Great, and Sad) and the associated factors (Home, Work, School, Friend, etc.). These moods and factors were self-reported by users. For example, when mentioning "home-related factors", it means that the users indicated that their reported mood is related to an activity related to their homes. The data were analyzed using IBM SPSS for Windows 27.0 [16]. To ensure more representative and accurate results, only complete records were included in the analysis. Records that had missing values were excluded. We first calculated descriptive analysis for moods and factors. Then we investigated the correlation between moods and factors, where a Chi-Square Test of Independence was used to examine the associations between moods and factors, while a Phi value was further computed to evaluate the strength of the associations.

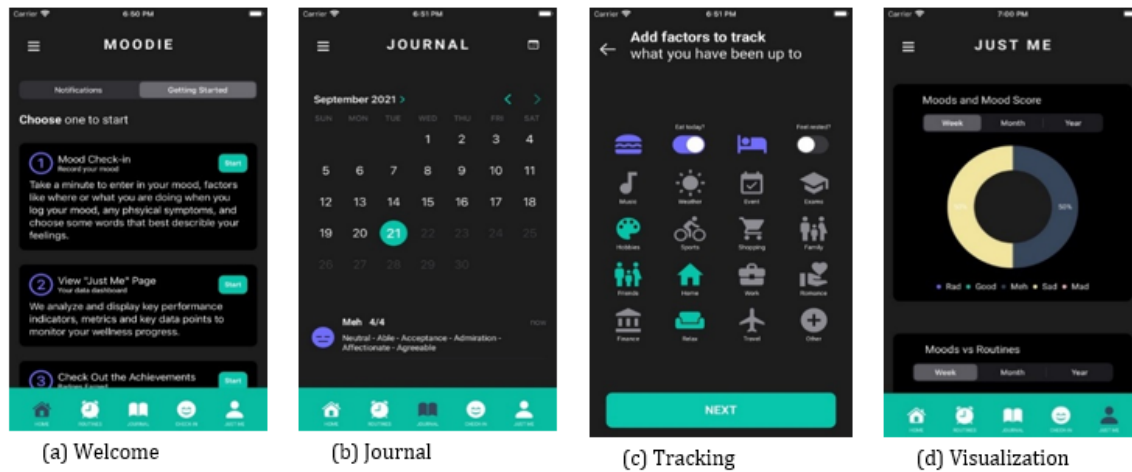


Figure 1: Moodie App Screenshots

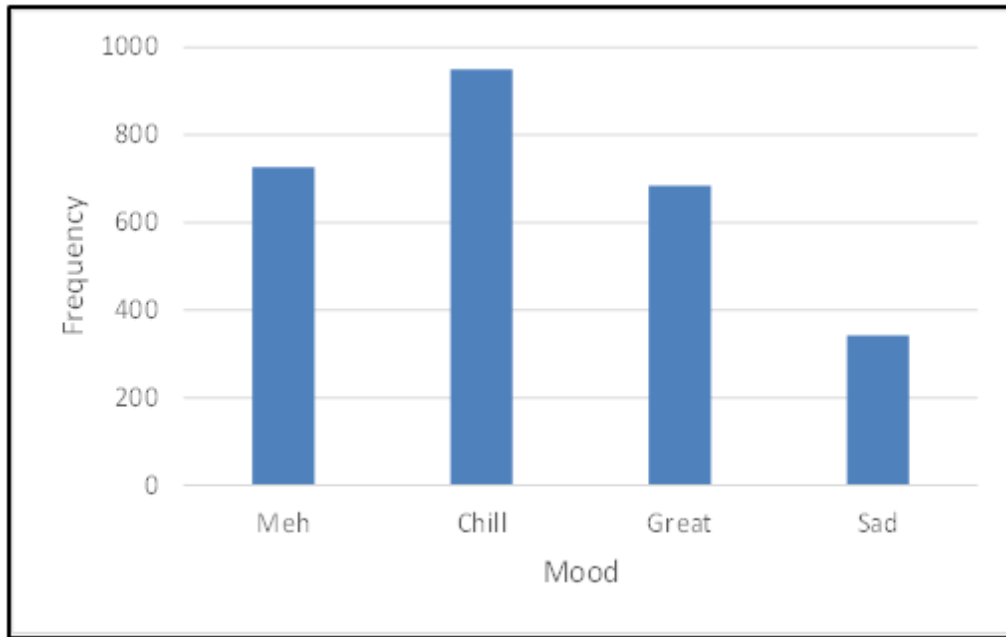


Figure 2: Distribution of Moods

5 EARLY RESULTS

This section represents the results based on the data collected in the first two years of using the *Moodie* app. First, it shows mood distribution based on the overall data (including all the moods and factors). Then, the section reports the most common factors that affect moods and the correlation between these factors and the four moods.

5.1 Mood Analysis

This section shows moods distribution for all users over the two years period. Figure 2 shows the number of occurrences for each mood. The figure shows that Chill was the most common mood; almost (35%) of all reported moods were Chill (i.e., most of the users used the app while they were in a calm state). Meh (or boredom) is the second most common mood followed by Great, while Sad emerged as the least common mood.

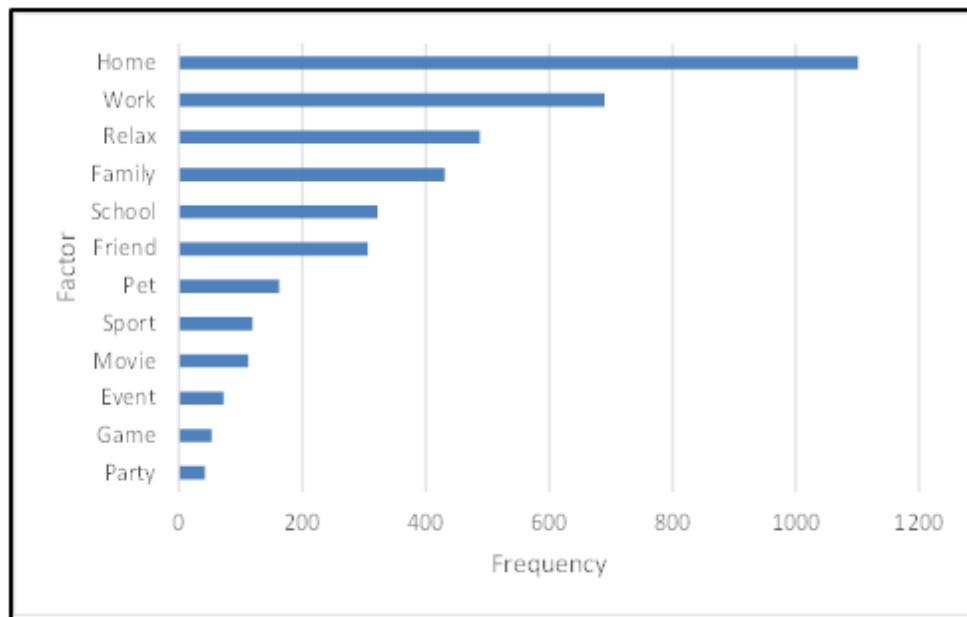


Figure 3: Distribution of Factors

5.2 Factors Analysis

In regard to the factors that affect moods, Figure 3 shows how many times the users associated each factor as a cause of their moods. As the chart depicts, Home-related activities (i.e., activities that are related or done while at home) or issues emerged as the most common factor among the twelve factors. Work- and Relaxation-related activities or issues were the second and third most common factors, respectively. This is followed by Family- and School-related activities or issues. On the other hand, Party, Game, and Event were the least common factors, where each factor was associated with less than 2% of the reported moods. These results indicate that users found Home- and Work-related activities as essential factors that may affect their moods either positively or negatively, but Work-related activities are not as common as Home-related activities.

5.2.1 Distribution of Factors Based on Time. People's feelings and moods are non-static, and can be affected by various aspects, including time [18]. Thus, we analyzed the data based on the time of the year to investigate the possible impact of season on the factors that affect moods. We divided the time based on seasons (Winter, Spring, Summer, and Fall). Figure 4 depicts the distribution of the top six factors based on the season. As the figure shows, home and work-related activities emerged as the most and second most common factors in all seasons. Relaxation emerged as the third most common factor in all seasons, except in summer, where Family emerged as the third most common factor. On the other hand, friends-related activities emerged as the least common factor in Winter, Spring, and Fall, whereas in Summer, school-related factors emerged as the least common factor among the top six factors. It is expected that school activities have less impact during the summer

as schools are closed at the beginning of the summer season for most people.

5.3 Correlation Between Moods and Factors

This section discusses how different moods are correlated with different factors or daily life activities. Table 1 summarizes the distribution of factors for each mood. It shows the number of times users associated each mood with every factor. A Chi-Square test was performed to assess the relationship between moods and factors. There was a significant relationship between the two variables, $\chi^2(33, N = 434) = 124.24, p = .001$. A Phi value was computed to evaluate the strength of the association. The strength of association between mood and activities is (.170), which indicates a strong association as it is greater than the threshold (.150) [1].

Figure 5 shows the relationship between moods and the factors for the whole dataset. It depicts the most common factors and shows how they are associated with each mood. Each bar shows the percentages of how often users associated their moods with each factor. The figure shows that Home- and Work-related activities emerged as the most and the second most common reasons for all moods. This is expected as people spend most of their time between home and work. On the other hand, the figure shows that friends-related aspects are the least common cause of Sad and Meh moods, while School is the least common cause of Great mood among the top six factors.

Figure 6 shows several interesting observations. Although Home-related factors emerged as the most common reason for all moods in the general dataset, this is not the case across various seasons of the year based on in-depth analysis. For instance, the School factor emerged as the most common factor associated with Great, Chill, and Meh moods in the spring season, and for Chill and Meh moods in the winter. In addition, Relaxation factor emerged as the

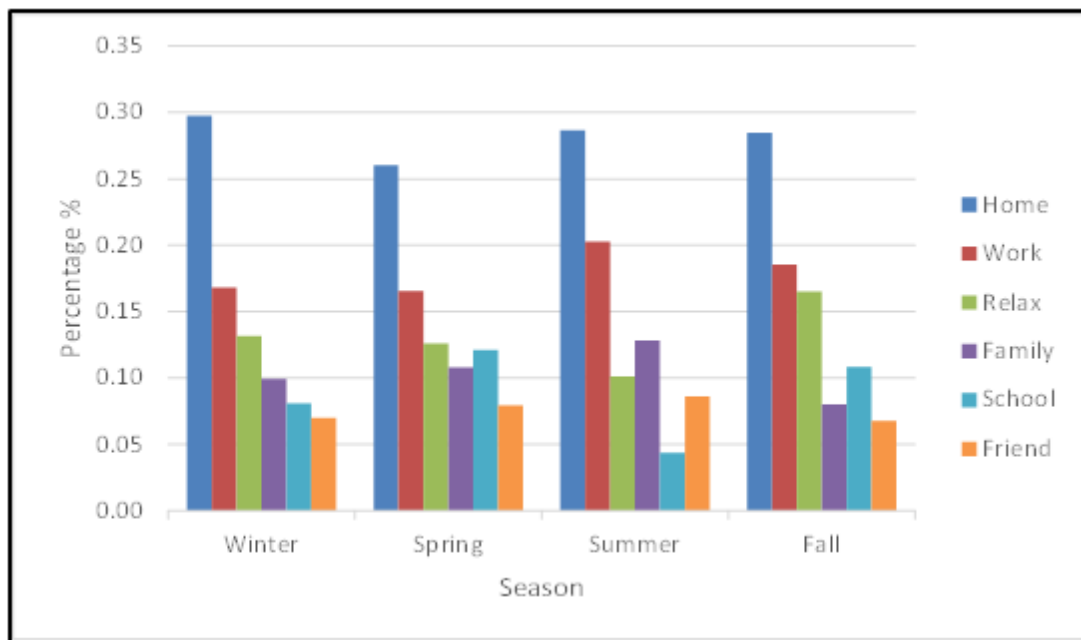


Figure 4: Distribution of Factors by Season (Top 6)

Table 1: Frequency of Factors and Moods

Mood	Work	School	Sport	Event	Party	Relax	Friend	Family	Home	Pet	Game	Movie
Meh	201	117	16	21	4	118	67	118	319	44	10	38
Chill	232	106	46	21	12	222	106	147	382	67	19	44
Great	181	54	51	20	21	109	102	108	257	34	16	23
Sad	76	45	6	10	5	39	31	58	143	17	8	7

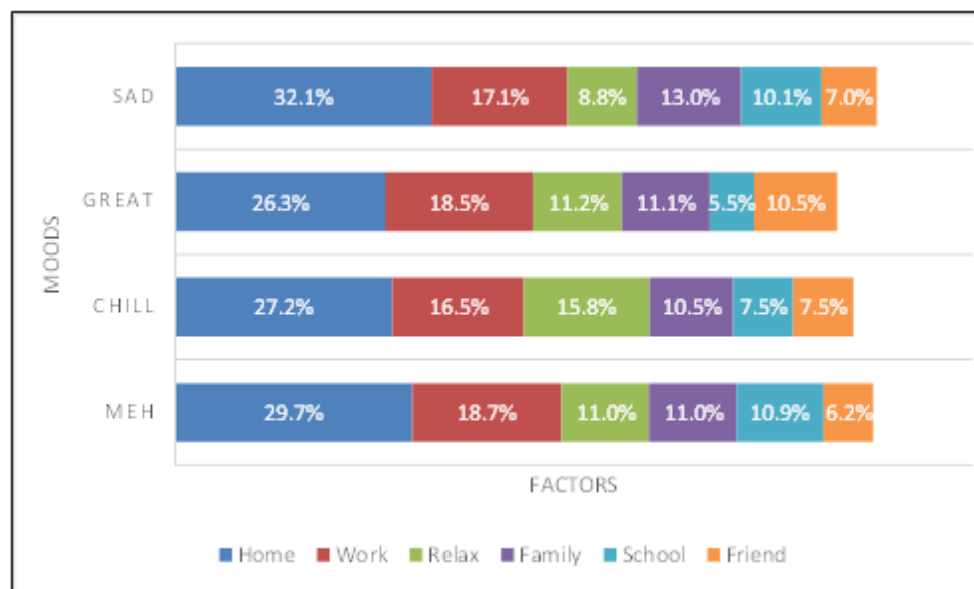


Figure 5: Correlation between Moods and Factors (Top 6)

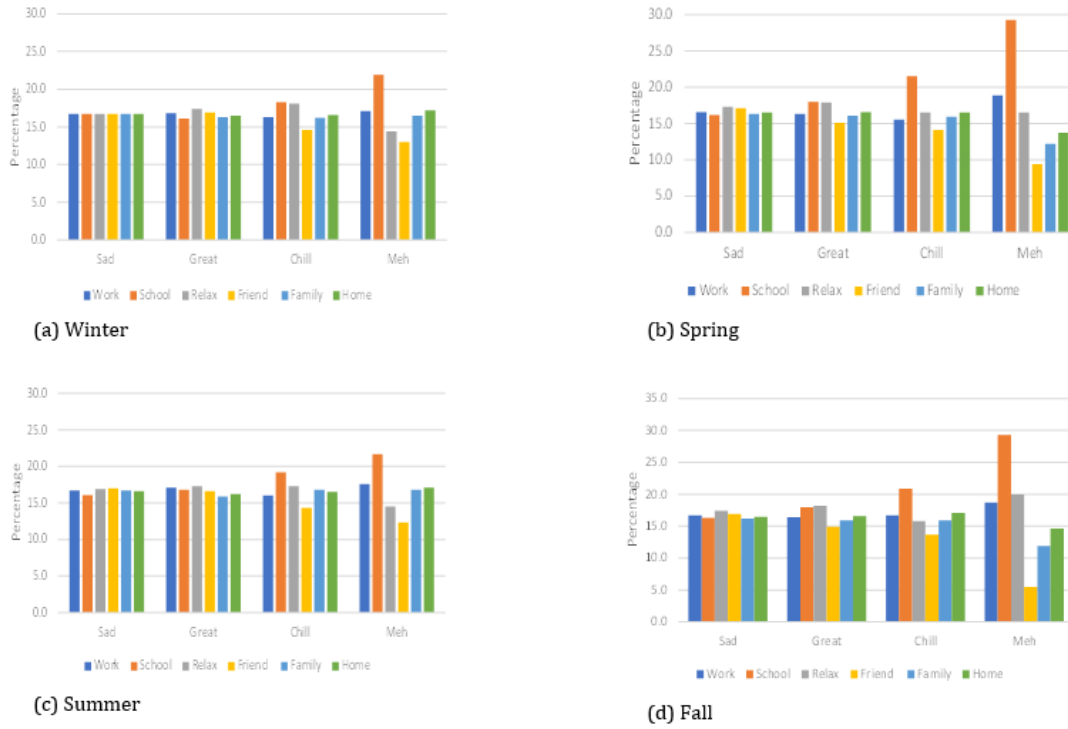


Figure 6: Correlation between Moods and Factors per season

most common cause of Great mood in winter, summer, and fall. This suggests that factors contributing to an individual's mood vary depending on the season of the year.

6 DISCUSSION AND FUTURE WORK

In the previous section, we presented the results obtained from our analysis. The results revealed interesting patterns that can inform the development of an adaptive AI-based mental health app. Following is a summary of the key observations and recommendations/guidelines for designing and researching mHealth apps:

- The results show that Home-related activities are the dominant factor that causes different moods – Meh, Chill, Great, and Sad. It emerged as the most common factor for the overall dataset. Also, work-related activities emerged as the second most common factor of all moods. This significance of Home and Work can be because people spend most of their time between home and work. Therefore, *mental health apps should emphasize home and work-related activities, triggers of various moods, and provide exercises that best fit the home and work environment and triggers*. For instance, apps may use serious games to train employees to take deep breaths regularly during working hours because deep breathing is an efficient way to reduce stress levels [21].
- School-related activities emerged as the least cause for Great and Chill moods among the most common factors (home, work, Relaxation, Family, School, and Friend), while Friends

emerged as the least common cause of Meh, Chill, and Sad moods in the overall dataset (see Figure 5). This can be attributed to the pandemic and the closure of most activities during the last two years. The precaution measures applied in most countries caused a closure for universities and schools for a long time when this data was collected. Also, the closure of indoor activities and the restrictions on social gatherings reduce friends' gatherings and activities. Therefore, **we conclude that mental health apps should take the context into consideration and should be adaptive to the current situation**. This recommendation is supported by previous research which found that apps that users' appreciate mental health apps that offer adaptive functionalities and allow users to personalize some features [2].

- Related to the previous point, we noticed that different factors may affect the same mood based on different times of the year. For instance, school-related activities were more common in the year 2021 compared to 2020, and they become less important in the summer while Friends become more important (see Figure 4). The reason behind these findings could be related to the closure of the schools in the summer and they were closed in 2020 because of the pandemic. During these closure times, students turn to spend more time with their friends. The family factor was also reported more times in the summer than in other seasons (see Figure 3). This may be because people spend more time with their families in the summer, and typically they go for vacations together. Previous research [2] found that people who suffer

from mental health issues often require assistance, including social support from family and friends, to cope with their conditions and avoid isolation. So, *we suggest that mental health apps should consider users' characteristics, time, and other contextual aspects* [25]. In particular, ***mental health apps should emphasize family and friend-related activities in the summer. Also, we recommend designers emphasize less school-related activities for treatment during the summer time.***

- Correlation analysis (Section 4.3) shows that moods and activities are significantly correlated. This observation is consistent with previous studies showing that moods and activities are related. For example, Triantafillou et al. [27] found that mood is related to sleep quality, and Grander et al. [13] found a correlation between mood and food choice. Furthermore, our results show differences in the associations between mood and activities for time of the year (Winter, Spring, Summer, and Fall). Accordingly, **a comprehensive study of the relationship between factors and moods is needed to enrich the development of adaptive mental health apps.**
- Great mood is less common than Chill and Meh among our users. This observation can be attributed to the fact that people use mental health apps mostly when they need support. So, users usually do not use these apps when they are in a Great mood. However, "Prevention is better than cure." Therefore, ***we recommend designing interventions that consider people who are in a good mood such that the app encourages them to stay in a good mood. Serious games*** [22], ***reminders and social influence techniques*** [3] ***can be used to achieve this goal.***

These are the key insight from the data analysis. However, there are many other insights that can be derived from our findings to inform mental health intervention design. As mentioned before, this paper is the first step towards developing an adaptive AI-based mental health app that is built based on an evidence-based approach and personalized to reflect various users' contexts. As future work, we will conduct a qualitative study to analyze users' journaling entries and collect more details about each factor and its corresponding activities. We will also develop machine learning models and evaluate their ability to predict an individual's emotional state using journaling logs and associated emotions from the Moodie app. Besides, we need to study the impact of the app on the other stakeholders, namely the therapists.

7 CONCLUSION

This paper presented results obtained from 434 users who used the *Moodie* app over two years period. First, it presented descriptive analysis, followed by an analysis of factors that affect mood and how these factors vary over time across various seasons of the year – Winter, Spring, Summer, and Fall. Then, the paper discussed the potential correlation between moods and factors. Finally, the paper provided insights and recommendations for designing mental health interventions. The results indicated that Chill (being in a calm state or experiencing non-judgmental reality acceptance) was the most common mood among users in the last two years. Home

and work-related activities emerged as the most common factors that impact moods. The results also showed a significant correlation between moods and factors. This work aims to inform the design of adaptive AI-based mental health app. In the future, we will use other methods to explore the correlation between moods and factors using qualitative data and machine learning models.

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